Object Oriented Programming (CSEG1044)

Unit 06: Capstone Project —Proposal

Cafeteria Order and Billing System

1. Problem Statement

Many college cafeterias rely on manual processes for taking orders, calculating bills, and maintaining sales records. This leads to long queues, frequent billing errors, and the inability to track daily sales efficiently. Staff struggle to manage an ever-changing menu without a reliable system, and customers face inconsistent service. The Cafeteria Order and Billing System aims to solve these problems by providing a desktop application that enables cafeteria operators to manage menu items, place and track customer orders, generate accurate bills, and view daily sales summaries — all through a clean and simple interface.

2. Target Users

Role	Responsibilities
Admin	Manages menu items (add/edit/remove), views daily sales summary, manages user accounts
Operator	Takes customer orders, selects items from menu, generates and prints bills, views order history

3. Core Entities (Java Classes)

Class	Purpose
MenuItem	Stores item name, category, price, and availability
Order	Represents a customer order with timestamp and status
OrderItem	Links a MenuItem to an Order with quantity
Customer	Stores customer name and contact information
Bill	Calculates total amount, taxes, and final payable amount
User	Represents Admin or Operator with login credentials

4. End-to-End Workflow

- Step 1: Operator logs in to the system using their credentials.
- Step 2: Operator selects items from the current menu displayed in the Swing UI.
- Step 3: System creates an Order and associated OrderItems for the selected items.
- Step 4: Operator confirms the order; system saves it to the database via JDBC.
- Step 5: System generates a Bill with itemised totals and the final amount payable.
- Step 6: Bill is displayed on screen and saved to order history.
- Step 7: Admin can later view daily sales summaries from the dashboard.

5. Version 1 Scope

Included in Version 1	Excluded from Version 1
Menu management (add/edit/remove items)	Payment gateway integration
Order placement via Swing UI	Kitchen automation / display system
Bill generation with item breakdown	Online or mobile ordering
Order history (view past orders)	Loyalty points or discount system
Daily sales summary for Admin	Multi-branch support

6. Initial Backlog

#	Task	Priority	Assigned To
1	Design MenuItem, Order, OrderItem, Customer, Bill, User classes	**High**	Member 1
2	Set up MySQL database schema with all tables	**High**	Member 2
3	Implement JDBC connection and DAO layer	**High**	Member 2
4	Build Login screen in Swing	**High**	Member 3
5	Build Menu Management UI (Admin)	**High**	Member 1
6	Build Order Placement UI (Operator)	**High**	Member 3
7	Implement Bill calculation logic (separate from UI)	**High**	Member 1
8	Build Bill display screen in Swing	**Medium**	Member 3
9	Implement Order History view	**Medium**	Member 4
10	Implement Daily Sales Summary (Admin)	**Medium**	Member 4
11	Testing and bug fixes	**High**	All Members

7. Why This Project Fits Unit 06

The Cafeteria Order and Billing System is an ideal project for Unit 06 because it directly demonstrates all three pillars required: Object-Oriented Programming, Java Swing, and JDBC. OOP is applied throughout the design — with encapsulation in entity classes such as MenuItem and Bill, inheritance for User roles (Admin and Operator), and clear separation of concerns between business logic and UI. Java Swing is used to build all user-facing screens including login, menu management, order placement, and billing. JDBC connects the application to a relational database, enabling persistent storage of orders, items, and bills. The project is realistic, achievable within the course timeline, and covers a complete end-to-end workflow that can be demonstrated clearly.

8. Group Members

#	Name	Roll No.	Role
1	Ritesh Singh	590017034	Member 1
2	Anurag Kumar	590011480	Member 2
3	Mohit Vats	590016048	Member 3
4	Bhavishya Pradhan	590016048	Member 4